Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 105616 | *Subject: Cryptography and Network Security* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) A polyalphabetic cipher can be classified as
- (i) Simple substitution
- (ii) Transposition cipher
- (iii) Multi-key substitution
- (iv) Stream cipher

(b) In transposition cipher, encryption is achieved by
- (i) Replacing characters with other characters
- (ii) Rearranging the order of characters
- (iii) Substituting binary digits
- (iv) Hiding message within images

(c) The key length of DES is
- (i) 56 bits
- (ii) 128 bits
- (iii) 64 bits
- (iv) 256 bits

(d) The avalanche effect in DES ensures
- (i) Minimal change in ciphertext for key variation
- (ii) Key independence
- (iii) Constant ciphertext length
- (iv) Large change in ciphertext for small plaintext/key change

(e) Which of the following statements about finite fields is true?
- (i) Every finite field has a prime number of elements
- (ii) Finite fields exist only for even p^n
- (iii) Every finite field has p^n elements, where p is prime
- (iv) All finite fields are of infinite order

(f) If gcd(a, n) = 1, the modular inverse of a (mod n) exists and can be found using
- (i) Fermat's theorem
- (ii) Chinese remainder theorem
- (iii) Discrete logarithm
- (iv) Extended Euclidean algorithm

(g) Which of the following is a stream cipher?
- (i) RC4
- (ii) DES
- (iii) AES
- (iv) Triple DES

(h) The security of Triple DES relies on
- (i) Using three distinct keys with DES encryption-decryption-encryption
- (ii) XOR operations
- (iii) Key scheduling randomness
- (iv) Increasing block size

(i) The Chinese Remainder Theorem helps in
- (i) Finding public key
- (ii) Generating random numbers
- (iii) Solving modular equations efficiently
- (iv) Detecting prime numbers

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(j) Given $p = 3, q = 11$, public key $e = 7$, and plaintext $M = 5$, find ciphertext C.

(i) 12 (iii) 24

(ii) 14 (iv) 10

Q.2 (a) Explain substitution and transposition cipher. Differentiate between both the techniques. [7]

(b) Draw a neat diagram OSI Model. Discuss the various services provided by the OSI Security Architecture and how they interrelate. [7]

Q.3 (a) Describe the structure and working process of DES with a neat diagram. [7]

(b) Explain the concepts of differential and linear cryptanalysis. Explain Block Cipher Principles. [7]

Q.4 (a) Explain Groups, Rings, and Fields. Use Euclid's Algorithm to find gcd(60, 24). [7]

(b) Explain the role of key distribution and random number generation in ensuring confidentiality in network security. [7]

Q.5 (a) Define AES. Explain the structure and major steps of the AES cipher. [7]

(b) Explain the concept of block cipher modes of operation such as ECB and CBC with diagram. [7]

Q.6 (a) Discuss in detail the RSA algorithm, including encryption, decryption, and security considerations. [7]

(b) Explain Principles of Public-Key Cryptosystems. Explain Fermat's and Euler's Theorems with example. [7]

Q.7 (a) Explain in detail the digital signature process. Explain the requirements for authentication in communication systems. [7]

(b) Define message authentication. Differentiate between Message Authentication Code (MAC) and Hash Function. [7]

Q.8 (a) Explain Playfair Cipher. A message "INFORMATION" is encrypted using a Playfair cipher with the keyword "KEYWORD". Construct the matrix and encrypt the message. [7]

(b) Define confusion and diffusion. What is the purpose of hash functions in authentication systems? [7]

Q.9 (a) Differentiate between symmetric and asymmetric encryption. What is the purpose of hash functions in authentication systems? [7]

(b) What is the difference between encryption and authentication? Find the multiplicative inverse of 7 mod 26. [7]

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